UNIT - I

Number Theory

Tutorial Sheet

Part A

1. If n is a multiple of 6, then n^3 is always divisible by __________.
2. If $\gcd(a, b) = d$, then $\gcd\left(\frac{a}{d}, \frac{b}{d}\right)$ is always __________.
3. The maximum possible value of $\gcd(a, b)$ when $a + b = 100$ is __________.
4. If $\gcd(a, b) = 1$, then there exist integers x and y such that $ax + by =$ __________.
5. If a and b are relatively prime, then $\gcd(a^2, b^3)$ is __________.
6. Every integer greater than 1 can be uniquely expressed as a product of __________.
7. The prime factorization of 360 is ____________________.
8. If p is a prime number and divides a^2, then p must also divide __________.
9. The remainder when 7^{100} is divided by 6 is __________.
10. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $(a + c) \equiv$ __________ $\pmod{m}$.
11. The multiplicative inverse of 3 modulo 7 is __________.
12. Euler's totient function $\phi(12)$ is equal to __________.
13. In RSA encryption, the public key is a pair (e, n), where e is the __________.
14. The decryption exponent d in RSA is the modular inverse of __________ modulo $\phi(n)$.

Part B

1. Prove that every integer greater than 1 has a unique prime factorization.
2. Find the prime factorization of 2520 and determine the number and sum of its divisors.
3. Prove that there are infinitely many prime numbers.
4. Solve the linear congruence $7x \equiv 1 \pmod{26}$.
5. Compute the last digit of 2^{100} using modular arithmetic.
6. Find all integer solutions to $3x \equiv 6 \pmod{9}$.

7. Use Euler's theorem to compute $3^{100} \pmod{101}$.
8. Given $p = 61$ and $q = 53$, compute n, $\phi(n)$, and suggest possible values for public and private keys.
9. By using the Euclidean algorithm, find the greatest common divisor d of 143 and 227 and then find integers x and y to satisfy $143x + 227y = d$. Also, show that x and y are not unique.
10. Find the greatest common divisor d of the numbers 272 and 1479 using Euclid's algorithm and then find integers x and y to satisfy $272x + 1479y = d$.
11. Find the remainder when 2^{23} is divided by 47.
12. Find the last digit in 7^{118}.
13. Find the last two digits in 36^{233}.
14. Find the remainder when $175 \times 113 \times 53$ is divided by 11.
15. Find the number of positive divisors and sum of all positive divisors of 1363.
16. Find the number of positive divisors and sum of all positive divisors of 8128.
17. Find the solutions of the linear congruence $11x \equiv 4 \pmod{25}$.
18. Find the solutions of the linear congruence $25x \equiv 15 \pmod{29}$.
19. Find the solutions of the linear congruence $6x \equiv 15 \pmod{21}$.
20. Find all distinct solutions of the linear congruence $60x \equiv 35 \pmod{625}$.
21. Find the multiplicative inverse of $113 \pmod{2036}$.
22. Encrypt the message **STOP** using RSA with key $(e, n) = (13, 2537)$ using the prime numbers 43 and 59.
23. Given the public key $(e, n) = (7, 85)$, encrypt plaintext **H C M**, where the alphabets $A, B, C, \ldots, X, Y, Z$ are assigned the numbers $3, 4, \ldots, 27, 28$. Give the ciphertext. Find the private key d.